

Searching for Axion-Like Particles at FCC-ee: Projected Sensitivity in the Invisible and Prompt-Resolved Channels at the Z Pole

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Abstract

We present projected sensitivity estimates for the production of axion-like particles (ALPs) at the proposed Future Circular Collider operating at the Z pole ($\sqrt{s} = 91.2 \text{ GeV}$, $\mathcal{L} = 150 \text{ ab}^{-1}$) via the associated production process $e^+e^- \rightarrow \gamma a$. Two detector signatures are analysed: the *invisible* channel, in which the ALP escapes the detector ($\ell_a > L_{\text{max}}$), and the *prompt-resolved* channel, in which the ALP decays to two separated photons inside the tracker ($\ell_a < L_{\text{min}}$, $\Delta\theta > \Delta\theta_{\text{res}}$). Signal events are generated with MADGRAPH5_AMC@NLO, processed through PYTHIA 8 for ALP decay and lifetime, and passed through a DELPHES fast simulation of the IDEA detector concept. SM backgrounds for both channels are simulated, binned, and normalised to the full target luminosity. Projected 90% CL exclusion contours are derived using a binned Asimov $\Delta\chi^2$ method, with analytic production and lifetime formulae corrected by detector-level signal scans. For the invisible channel, the analysis reaches $g_{a\gamma\gamma} \approx 5.5 \times 10^{-7} \text{ GeV}^{-1}$ over the ALP mass range 0.01–0.92 GeV. The prompt-resolved channel extends sensitivity to $g_{a\gamma\gamma} \sim 10^{-5} \text{--} 3 \times 10^{-4} \text{ GeV}^{-1}$ for $m_a \in [0.61, 80] \text{ GeV}$. Both projected contours surpass current collider bounds by one to two orders of magnitude in coupling at comparable masses. The analysis machinery is validated against the published Belle II ALP exclusion boundary, reproducing it to within $|\log_{10}(g_{\text{closure}}/g_{\text{published}})| \leq 7.6 \times 10^{-3}$. Additional validation gates check the MadGraph cross section, the ALP width and Pythia lifetime convention, the Delphes output, and the reconstructed ALP invariant-mass or recoil-energy observable at every detector-level signal point.

1 Introduction

The Peccei–Quinn (PQ) mechanism, proposed to explain the absence of strong CP violation [1–3], predicts the existence of a pseudo-Nambu–Goldstone boson, the axion. The simplest requirement for the PQ mechanism to resolve the strong CP problem constrains the axion mass and its couplings to a narrow band in the coupling–mass plane—the so-called QCD axion line. However, extensions of this framework that decouple the mass from the coupling give rise to *axion-like particles* (ALPs): pseudo-scalars that couple to Standard Model (SM) particles with arbitrary mass–coupling combinations [4–6].

The ALP–photon coupling,

$$\mathcal{L} \supset -\frac{1}{4}g_{a\gamma\gamma}aF_{\mu\nu}\tilde{F}^{\mu\nu}, \quad (1)$$

is particularly well-motivated. It arises generically from the anomalous coupling of any axion-like field to the electromagnetic field strength $F_{\mu\nu}$, and is bounded across many orders of magnitude in mass by astrophysical, cosmological, and laboratory experiments [12]. In the mass range 10 MeV–80 GeV relevant to collider searches, the most stringent direct constraints come from e^+e^- collider experiments (LEP, Belle II, BaBar, BESIII) and beam-dump experiments, and leave significant regions of parameter space unexplored at couplings $g_{a\gamma\gamma} \lesssim 10^{-4} \text{ GeV}^{-1}$.

The Future Circular Collider (FCC-ee) is a proposed e^+e^- machine that would operate at several centre-of-mass energies from the Z pole ($\sqrt{s} \approx 91.2 \text{ GeV}$) up to the $t\bar{t}$ threshold [8, 9]. At the Z pole, the baseline luminosity target of 150 ab^{-1} —corresponding to approximately 5×10^{12} Z bosons—provides an extraordinarily clean and prolific environment for precision measurements and rare-process searches. For photophilic ALPs, the associated production cross section $\sigma(e^+e^- \rightarrow \gamma a) \propto g_{a\gamma\gamma}^2(1 - m_a^2/s)^3$ grows quadratically with the coupling and benefits enormously from the high luminosity, making FCC-ee a uniquely powerful probe. The Z-pole run is chosen as the baseline deliverable for three practical and physics reasons. First, it is the FCC-ee stage with the largest expected integrated luminosity, so it gives the strongest statistics-driven reach for the single-coupling process studied here. Second, the fixed centre-of-mass energy gives a simple mono-energetic recoil photon, $E_\gamma = (s - m_a^2)/(2\sqrt{s})$, making the invisible and prompt-resolved analyses especially transparent. Third, the Z-pole overlaps the mass range where existing collider constraints from Belle II, BaBar, BESIII, LEP, and the LHC leave open regions that can be tested in a clean e^+e^- environment. Other FCC-ee stages are scientifically interesting, but each stage has a different luminosity, background composition, recoil-energy range, and detector acceptance optimisation. They are therefore treated as extensions rather than part of the controlled final deliverable.

This paper presents projected ALP sensitivity estimates at the FCC-ee Z pole, focusing on the two experimentally cleanest signatures: the invisible channel (ALP escapes) and

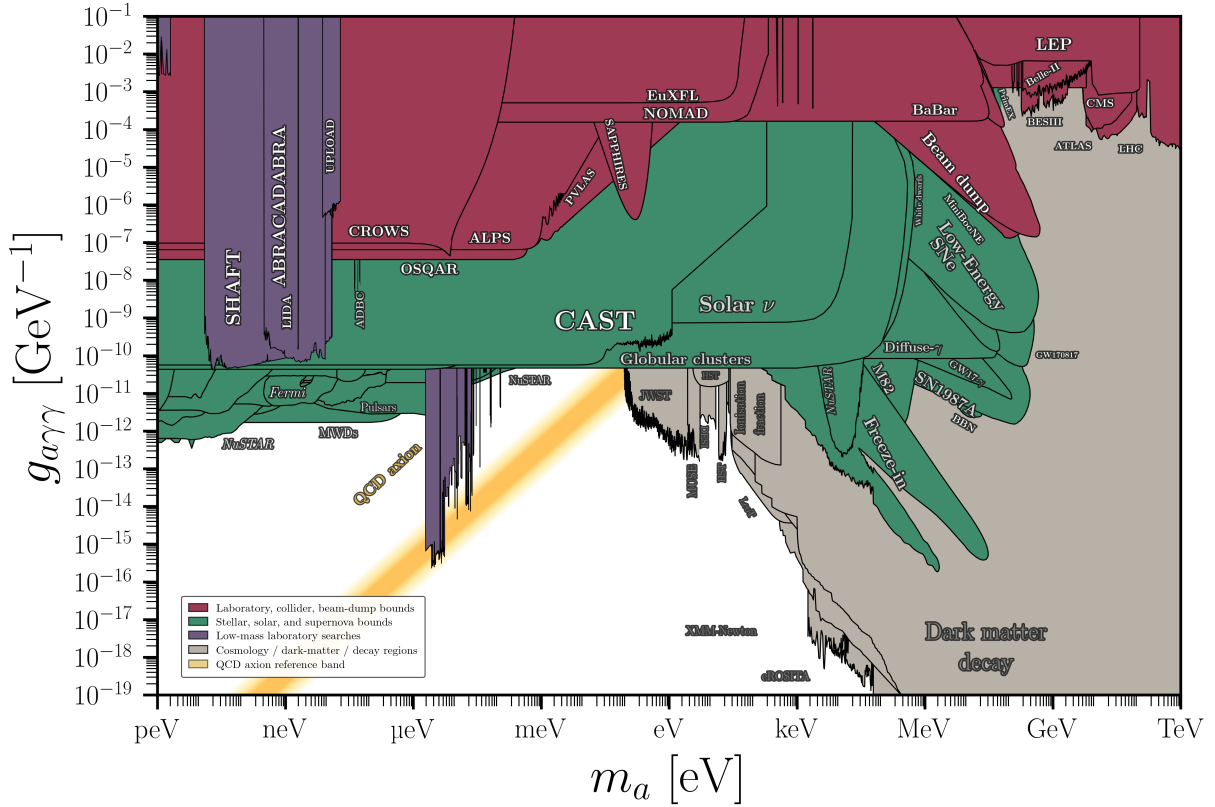


Figure 1: Current ALP-photon constraint landscape used as context for this project, produced from the pinned AxionLimits repository. Solid coloured regions show existing laboratory, astrophysical, cosmological, and dark-matter-motivated constraints or reference regions. The FCC-ee projections are deliberately omitted here and are introduced later as the project result.

the prompt-resolved channel (ALP decays to two separated photons). The analysis uses a complete simulation chain from matrix-element generation to fast detector response, validated against published experimental results.

A central goal of the project is not only to draw an ALP sensitivity curve, but to build a reusable ALP simulation and analysis pipeline. This matters because an ALP search is not determined by the production formula alone. The same point in the $(m_a, g_{a\gamma\gamma})$ plane can look like missing energy, three photons, two merged photon clusters, or a displaced decay depending on the ALP lifetime and boost. The detector response then decides how many of those ideal events are actually reconstructed, while Standard Model processes decide how difficult the signal is to distinguish from ordinary events. A pipeline that connects theory, event generation, detector simulation, background modelling, and limit setting is therefore needed if the final projection is to mean more than an idealised rate estimate. It also makes the analysis extensible: the same structure can later be reused for other FCC-ee energies, different detector assumptions, displaced or merged channels, or additional ALP operators.

Paper structure. Section 2 reviews the relevant ALP physics. Section 3 describes the simulation and detector modelling. Section 4 presents the signal classification, background estimation, and limit-setting method. Section 5 reports the projected exclusion contours. Section 6 discusses the physics reach, limitations, and outlook.

2 Theory

2.1 Analytic model for photophilic ALPs

This analysis assumes a single effective operator, Eq. (1), with all direct ALP couplings to fermions, gluons, W bosons, and Z bosons set to zero at tree level. The only model parameters scanned are therefore the ALP mass m_a and the photon coupling $g_{a\gamma\gamma}$. To calculate any process we need to compute the amplitude of the interaction. This is derived from the Lagrangian of the interaction (1). In this benchmark the total width is taken to be

$$\Gamma_{\text{tot}} \simeq \Gamma(a \rightarrow \gamma\gamma), \quad (2)$$

because $a \rightarrow \gamma\gamma$ is the only open tree-level decay channel in the implemented model. Loop-induced fermion modes are neglected consistently with the photophilic benchmark and with the UFO parameter choice used in the simulation.

2.2 ALP–photon vertex and associated production

The collider event is easiest to read in the same order as it happens. First, the incoming electron and positron annihilate through the ordinary QED vertex into a virtual photon,

$$e^+e^- \rightarrow \gamma^*(q), \quad (3)$$

where $q = p_{e^-} + p_{e^+}$ is the four-momentum carried by the virtual photon. This photon is *off shell*: a real photon must satisfy $q^2 = 0$, but here $q^2 = s$, the squared centre-of-mass energy of the collision. The virtual photon then uses the ALP–photon interaction to produce a real recoil photon and an ALP,

$$\gamma^*(q) \rightarrow \gamma(k) a(p_a). \quad (4)$$

The full associated-production process is therefore

$$e^+e^- \rightarrow \gamma a, \quad (5)$$

with the real photon providing the mono-energetic recoil tag. Photon-fusion production $e^+e^- \rightarrow e^+e^-a$ can have a larger inclusive rate, but it places the scattered electrons in

the forward beampipe region and is much harder to separate from radiative Bhabha and beam-induced backgrounds. The analytic and Monte Carlo pipeline therefore targets the recoil-photon topology.

The same ALP-photon vertex controls both this production process and the later ALP decay. Using

$$\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \quad \epsilon^{0123} = +1, \quad (6)$$

the interaction term gives the two photon vertex

$$V^{\mu\nu}(k_1, k_2) = -i g_{a\gamma\gamma} \epsilon^{\mu\nu\alpha\beta} k_{1\alpha} k_{2\beta}, \quad (7)$$

where both photon four-momenta, k_1 and k_2 , are taken incoming to the vertex. Here $F_{\mu\nu}$ is the electromagnetic field-strength tensor, while $\tilde{F}^{\mu\nu}$ is its dual. The object $\epsilon^{\mu\nu\rho\sigma}$ is the four-dimensional Levi-Civita symbol. It is a completely antisymmetric object which encodes the CP-odd nature of the ALP-photon interaction. It equals +1 for the ordered indices (0, 1, 2, 3), changes sign whenever two indices are exchanged, and vanishes whenever two indices are the same. The Greek indices μ, ν, α, β are Lorentz indices, and repeated indices are summed over according to the Einstein convention. In production, one of these photon legs is the off-shell $\gamma^*(q)$, and the other is the recoil photon $\gamma_{\text{recoil}}(k)$.

There is one sign that is easy to miss when using this vertex in production. Equation (7) is written with both photon momenta incoming to the ALP vertex. In the physical process $e^+e^- \rightarrow a\gamma$, however, the recoil photon is outgoing. Therefore the momentum that must be inserted into the all-incoming Feynman rule is $-k$, not k . If the virtual photon carries incoming momentum q and Lorentz index ν , while the outgoing recoil photon has momentum k and Lorentz index ρ , then

$$V^{\nu\rho}(q, -k) = -i g_{a\gamma\gamma} \epsilon^{\nu\rho\alpha\beta} q_\alpha (-k)_\beta = i g_{a\gamma\gamma} \epsilon^{\nu\rho\alpha\beta} q_\alpha k_\beta. \quad (8)$$

This is why the production amplitude contains $i g_{a\gamma\gamma} \epsilon^{\nu\rho\alpha\beta} q_\alpha k_\beta$ instead of the original all-incoming $-i g_{a\gamma\gamma} \epsilon^{\mu\nu\alpha\beta} k_{1,\alpha} k_{2,\beta}$. The remaining overall phase is convention-dependent and drops out of $|\mathcal{M}|^2$.

The amplitude for $e^+e^- \rightarrow a\gamma$ is therefore obtained by multiplying the ordinary QED current, the virtual-photon propagator, the ALP-photon vertex, and the polarization vector of the outgoing photon:

$$\begin{aligned} \mathcal{M} &= \underbrace{\bar{v}(p_{e^+})(-ie\gamma^\mu)u(p_{e^-})}_{e^+e^-\gamma^* \text{ current}} \underbrace{\left(\frac{-ig_{\mu\nu}}{s}\right)}_{\gamma^* \text{ propagator}} \underbrace{(ig_{a\gamma\gamma}\epsilon^{\nu\rho\alpha\beta}q_\alpha k_\beta)_{\gamma^*\gamma a \text{ vertex}}} \underbrace{\varepsilon_\rho^*(k)}_{\text{outgoing } \gamma \text{ polarization}} \\ &= \frac{eg_{a\gamma\gamma}}{s} [\bar{v}(p_{e^+})\gamma^\mu u(p_{e^-})] \epsilon_{\mu\nu\alpha\beta} q^\alpha k^\beta \varepsilon^{*\nu}(k). \end{aligned} \quad (9)$$

In the second line the Lorentz indices have been contracted and the overall phase has been dropped, since only $|\mathcal{M}|^2$ enters the cross section. The factors in this expression have direct diagrammatic meanings. The electric charge e comes from the ordinary QED vertex where the electron-positron pair annihilates into a virtual photon. The spinors $u(p_{e-})$ and $\bar{v}(p_{e+})$ describe the incoming electron and positron, and γ^μ is a Dirac matrix. The factor $1/s$ is the propagator of the intermediate virtual photon after contracting with the photon-current indices. The factor $g_{a\gamma\gamma}\epsilon_{\mu\nu\alpha\beta}q^\alpha k^\beta$ is the ALP-photon vertex from Eq. (7); q is the virtual-photon momentum and k is the outgoing real-photon momentum. Finally, $\varepsilon^{*\nu}(k)$ is the polarization vector of the observed outgoing photon.

2.3 Two-body production kinematics and cross section

In the centre-of-mass frame the total four-momentum is $P^\mu = p_{e-}^\mu + p_{e+}^\mu = (\sqrt{s}, 0, 0, 0)$. Momentum conservation gives $P^\mu = p_a^\mu + k^\mu$, with $k^2 = 0$ for the recoil photon and $p_a^2 = m_a^2$ for the ALP. Squaring $P - k = p_a$ gives

$$(P - k)^2 = m_a^2 \quad \Rightarrow \quad s - 2P \cdot k = m_a^2. \quad (10)$$

Here p_{e-} and p_{e+} are the incoming electron and positron four-momenta, p_a is the outgoing ALP four-momentum, and k is the outgoing recoil-photon four-momentum. The invariant $s = P^2 = (p_{e-} + p_{e+})^2$ is the squared centre-of-mass energy, so $\sqrt{s}$ is the total collision energy. Since $P \cdot k = \sqrt{s} E_\gamma$ in the centre-of-mass frame, the recoil photon energy is fixed:

$$E_{\gamma,\text{recoil}} = \frac{s - m_a^2}{2\sqrt{s}}. \quad (11)$$

The ALP energy and common three-momentum magnitude are therefore

$$E_a = \sqrt{s} - E_\gamma = \frac{s + m_a^2}{2\sqrt{s}}, \quad |\mathbf{p}_a| = |\mathbf{k}| = \frac{s - m_a^2}{2\sqrt{s}}. \quad (12)$$

These relations are the source of the mono-energetic recoil-photon signature. They are purely kinematic: at fixed $\sqrt{s}$ and m_a , a two-body final state has no freedom to choose a different recoil-photon energy before detector smearing or initial-state radiation. They also give the ALP boost,

$$\gamma_a = \frac{E_a}{m_a}, \quad \beta_a \gamma_a = \frac{|\mathbf{p}_a|}{m_a}. \quad (13)$$

Averaging over the two electron and two positron spin states and summing over the final photon polarisation gives, in the limit $m_e^2 \ll s$,

$$\overline{|\mathcal{M}|^2} = \frac{e^2 g_{a\gamma\gamma}^2 s}{4} (1 + \cos^2 \theta_{\text{CM}}) \left(1 - \frac{m_a^2}{s}\right)^2, \quad (14)$$

where θ_{CM} is the recoil-photon polar angle relative to the incoming electron direction. The spin average appears because the FCC-ee beams are treated as unpolarized: the calculation averages over the four possible initial spin configurations and sums over the unobserved final photon polarization. The factor $(1 + \cos^2 \theta_{\text{CM}})$ gives the angular shape, while $(1 - m_a^2/s)^2$ reflects the momentum dependence of the pseudoscalar vertex.

The two-body phase-space formula for unpolarised $2 \rightarrow 2$ scattering is

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\overline{\mathcal{M}}|^2, \quad (15)$$

with $|\mathbf{p}_i| = \sqrt{s}/2$ and $|\mathbf{p}_f| = (s - m_a^2)/(2\sqrt{s})$. In this expression $d\Omega$ is the solid-angle element of the outgoing photon, $|\mathbf{p}_i|$ is the magnitude of the incoming electron or positron three-momentum in the centre-of-mass frame, and $|\mathbf{p}_f|$ is the magnitude of either final-state momentum. The ratio $|\mathbf{p}_f|/|\mathbf{p}_i|$ is the two-body phase-space suppression: if m_a approaches $\sqrt{s}$, there is little momentum left for the final state. Integrating over the azimuthal angle then gives

$$\frac{d\sigma}{d\cos\theta_{\text{CM}}} = \frac{\alpha g_{a\gamma\gamma}^2}{32} (1 + \cos^2 \theta_{\text{CM}}) \left(1 - \frac{m_a^2}{s}\right)^3. \quad (16)$$

The third power of $(1 - m_a^2/s)$ has a useful interpretation: one power comes from the available two-body phase space and two powers come from the pseudoscalar momentum dependence of the $a\gamma\gamma$ vertex. Finally,

$$\int_{-1}^1 (1 + \cos^2 \theta) d\cos\theta = \frac{8}{3}, \quad (17)$$

so the total cross section over the full angular range is

$$\sigma(e^+e^- \rightarrow \gamma a) = \frac{\alpha g_{a\gamma\gamma}^2}{12} \left(1 - \frac{m_a^2}{s}\right)^3. \quad (18)$$

At $\sqrt{s} = 91.2 \text{ GeV}$ and $g_{a\gamma\gamma} = 10^{-3} \text{ GeV}^{-1}$, Eq. (18) gives $\sigma \approx 5.5 \times 10^{-5} \text{ pb}$; at $g_{a\gamma\gamma} = 10^{-6} \text{ GeV}^{-1}$ it falls to $5.5 \times 10^{-11} \text{ pb}$. The enormous FCC-ee Z-pole luminosity ($150 \text{ ab}^{-1} = 1.5 \times 10^8 \text{ pb}^{-1}$) compensates for small couplings and is what makes sensitivity below 10^{-6} GeV^{-1} possible.

2.4 ALP decay to two photons

After the ALP is produced, it can decay through the same $a\gamma\gamma$ vertex into two photons,

$$a(p) \rightarrow \gamma(k_1)\gamma(k_2). \quad (19)$$

The decay amplitude is

$$\mathcal{M} = -g_{a\gamma\gamma} \epsilon^{\mu\nu\alpha\beta} k_{1\alpha} k_{2\beta} \varepsilon_{1\mu}^* \varepsilon_{2\nu}^*. \quad (20)$$

In this equation p is the ALP four-momentum, while k_1 and k_2 are the four-momenta of the two decay photons. $\varepsilon_{1\mu}$ and $\varepsilon_{2\nu}$ are the photon polarization four-vectors. The star means complex conjugation, as appropriate for outgoing external particles. These polarization vectors describe the two physical transverse polarization states of a photon. They should not be confused with the Levi-Civita symbol $\epsilon^{\mu\nu\alpha\beta}$; the different symbol ε^μ is used here to keep the two concepts separate.

The detector does not measure the photon polarization in this analysis. Therefore the observable decay probability must include all allowed final polarization states. This is why we compute $\sum_{\text{pol}} |\mathcal{M}|^2$ rather than a single fixed-polarization amplitude. Because the amplitude is gauge invariant, terms proportional to a photon momentum vanish by the Ward identity, so the physical polarization sums can be replaced by the compact covariant expression $\sum_\lambda \varepsilon_\mu^{(\lambda)*} \varepsilon_{\mu'}^{(\lambda)} \rightarrow -g_{\mu\mu'}$. With the mostly-minus metric, this gives

$$\begin{aligned} \sum_{\text{pol}} |\mathcal{M}|^2 &= g_{a\gamma\gamma}^2 \epsilon^{\mu\nu\alpha\beta} \epsilon_{\mu\nu}^{\alpha'\beta'} k_{1\alpha} k_{2\beta} k_{1\alpha'} k_{2\beta'} \\ &= -2g_{a\gamma\gamma}^2 [k_1^2 k_2^2 - (k_1 \cdot k_2)^2]. \end{aligned} \quad (21)$$

The first line follows from multiplying Eq. (20) by its complex conjugate and summing over the two photon polarizations. The second line uses the standard Levi-Civita contraction identity

$$\epsilon^{\mu\nu\alpha\beta} \epsilon_{\mu\nu}^{\alpha'\beta'} = -2 \left(g^{\alpha\alpha'} g^{\beta\beta'} - g^{\alpha\beta'} g^{\beta\alpha'} \right). \quad (22)$$

The photons are on shell and massless, so $k_1^2 = k_2^2 = 0$. Momentum conservation gives $p = k_1 + k_2$, and because the ALP is on shell, $p^2 = m_a^2$. Therefore $m_a^2 = (k_1 + k_2)^2 = 2k_1 \cdot k_2$,

so

$$\sum_{\text{pol}} |\mathcal{M}|^2 = 2g_{a\gamma\gamma}^2 (k_1 \cdot k_2)^2 = \frac{g_{a\gamma\gamma}^2 m_a^4}{2}. \quad (23)$$

For correctly paired photons from the ALP decay,

$$M_{\gamma\gamma}^2 = (k_1 + k_2)^2 = 2E_1 E_2 (1 - \cos \theta_{12}) = m_a^2. \quad (24)$$

Here E_1 and E_2 are the measured photon energies and θ_{12} is the opening angle between the two photons. This invariant mass is independent of the lab-frame boost of the ALP, so correctly paired decay photons form a peak at $M_{\gamma\gamma} = m_a$. That peak is the analysis observable in the prompt-resolved channel.

For a two-body decay into identical massless particles,

$$\Gamma = \frac{1}{2m_a} \frac{1}{2!} \int d\Phi_2 \sum_{\text{pol}} |\mathcal{M}|^2, \quad \int d\Phi_2 = \frac{1}{8\pi}. \quad (25)$$

This formula is Fermi's golden rule written in relativistic form. The factor $1/(2m_a)$ is the flux normalization for one decaying particle of mass m_a . The factor $1/2!$ is essential because the two final photons are identical: without it, the same physical final state would be counted twice when the labels 1 and 2 are exchanged. The object $d\Phi_2$ is the Lorentz-invariant two-body phase-space element; for two massless particles integrated over all directions it equals $1/(8\pi)$. Equivalently, $\Gamma = (1/2!) |\mathbf{k}| |\mathcal{M}|^2 / (8\pi m_a^2)$ with $|\mathbf{k}| = m_a/2$, where $|\mathbf{k}|$ is the magnitude of either photon three-momentum in the ALP rest frame. Substituting Eq. (23) gives the width used throughout the project:

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}. \quad (26)$$

The proper lifetime and proper decay length are then

$$\tau_a = \frac{1}{\Gamma_a}, \quad c\tau_a = \frac{\hbar c}{\Gamma_a}, \quad (27)$$

with $\hbar c = 1.973269804 \times 10^{-16}$ GeV m. The lifetime τ_a is the mean time before decay in the ALP rest frame. Multiplying by c converts this time into the mean proper distance travelled in that rest frame. In the lab frame the decay length is increased by the boost,

$$\ell_a = \beta_a \gamma_a c\tau_a = \frac{|\mathbf{p}_a|}{m_a} \frac{\hbar c}{\Gamma(a \rightarrow \gamma\gamma)}. \quad (28)$$

where $\beta_a = v_a/c$, $\gamma_a = E_a/m_a$, and $\beta_a \gamma_a = |\mathbf{p}_a|/m_a$. This is the quantity that determines whether the ALP decays before the tracker, inside the detector, or outside the calorimeter. Equations (26) and (28) show that ℓ_a decreases with increasing $g_{a\gamma\gamma}$ and increasing m_a : heavy, strongly coupled ALPs decay promptly, while light, weakly coupled ALPs are long-lived and escape the detector.

2.5 Decay probabilities and photon separation

The probability density for an ALP to decay after travelling a lab-frame distance L is exponential,

$$f(L) = \frac{1}{\ell_a} e^{-L/\ell_a}. \quad (29)$$

This is the usual decay law written in distance rather than time. The parameter ℓ_a is the mean lab-frame decay distance from Eq. (28); a larger ℓ_a means that the ALP is more likely to survive to the calorimeter or escape the detector. Integrating this density over

the relevant detector length intervals gives the probabilities used in the analytic scan:

$$P(L > L_{\max}) = e^{-L_{\max}/\ell_a}, \quad (30)$$

$$P(L < L_{\min}) = 1 - e^{-L_{\min}/\ell_a}, \quad (31)$$

$$P(L_{\min} < L < L_{\max}) = e^{-L_{\min}/\ell_a} - e^{-L_{\max}/\ell_a}. \quad (32)$$

The first line is the survival probability to the outer detector radius $L_{\max}$ and defines the invisible signature. The second is the probability to decay before the prompt-distance scale $L_{\min}$. The third is the probability to decay between those two scales, producing a displaced signature. These probabilities define the invisible, prompt, and displaced regions of the grid.

The angular resolution boundary follows from boosting the isotropic $a \rightarrow \gamma\gamma$ decay. For an ALP moving along the z axis, the two photons are emitted back-to-back in the ALP rest frame. The smallest lab-frame opening angle occurs when the rest-frame photons are emitted transverse to the boost, for which

$$\Delta\theta_{\gamma\gamma}^{\min} = 2 \arctan\left(\frac{1}{\gamma_a\beta_a}\right) = 2 \arctan\left(\frac{m_a}{|\mathbf{p}_a|}\right). \quad (33)$$

The physical reason is simple: a highly boosted ALP carries the two decay photons forward in almost the same direction, so the two electromagnetic showers can merge in the calorimeter. Equation (33) uses the exact boost factor through $\gamma_a\beta_a = |\mathbf{p}_a|/m_a$. For light ALPs, $|\mathbf{p}_a| \simeq \sqrt{s}/2$ and $\gamma_a \simeq \sqrt{s}/(2m_a)$, so

$$\Delta\theta_{\gamma\gamma}^{\min} \simeq \frac{2}{\gamma_a} \simeq \frac{4m_a}{\sqrt{s}}. \quad (34)$$

The prompt-resolved channel requires this opening angle to be larger than the detector angular resolution, $\Delta\theta_{\text{res}}$. If $\Delta\theta_{\gamma\gamma}^{\min} < \Delta\theta_{\text{res}}$, the two photons are treated as merged even if the ALP decays promptly.

2.6 Signal signatures

The ALP phenomenology at the detector level is determined by two quantities: the decay length ℓ_a and the minimum diphoton opening angle $\Delta\theta_{\min}$ of the ALP decay products. The opening-angle criterion uses the boosted two-body result derived in Eq. (34). We classify each point $(m_a, g_{a\gamma\gamma})$ in the parameter space into four exclusive signatures:

- **Invisible** ($\ell_a > L_{\max}$): the ALP escapes before decaying. The signature is a single photon with missing energy, $e^+e^- \rightarrow \gamma + \cancel{E}$.
- **Prompt-resolved** ($\ell_a < L_{\min}$ and $\Delta\theta_{\min} \geq \Delta\theta_{\text{res}}$): the ALP decays immediately to two resolvable photons, $e^+e^- \rightarrow \gamma(\rightarrow \gamma\gamma)$. The three-photon final state carries

an invariant-mass peak at m_a .

- **Displaced-resolved** ($L_{\min} \leq \ell_a \leq L_{\max}$): the ALP decays inside the detector volume but away from the interaction point.
- **Merged** ($\ell_a \leq L_{\max}$ and $\Delta\theta_{\min} < \Delta\theta_{\text{res}}$): the ALP decay photons are not angularly resolved; the signature resembles a single broad photon cluster.

The invisible and prompt-resolved signatures are the primary targets of this analysis. Displaced and merged signatures require dedicated reconstruction and background models, so they are classified but not included as exclusion contours in the present study.

3 Simulation and Detector Modelling

3.1 From formulas to a collider search

The simplest way to understand the analysis is as a counting problem repeated many times across the $(m_a, g_{a\gamma\gamma})$ plane. At a collider, a *signal* is the process we are trying to discover or exclude. In this project the signal is

$$e^+e^- \rightarrow \gamma a, \quad a \rightarrow \gamma\gamma \quad \text{or} \quad a \text{ escapes before decaying.} \quad (35)$$

A *background* is a Standard Model process that can look similar in the detector even though no ALP was produced. A *yield* is the expected number of events of a given type in the dataset. In collider physics the first estimate of a yield is

$$N = \sigma \mathcal{L}, \quad (36)$$

where σ is the cross section and $\mathcal{L}$ is the integrated luminosity. The cross section measures how likely the process is to occur in a collision. The integrated luminosity measures the size of the dataset: larger luminosity means more collisions and therefore more chances to produce rare processes. The units are chosen so that if σ is in pb and $\mathcal{L}$ is in pb^{-1} , their product is a pure event count. The FCC-ee Z-pole dataset used here is $\mathcal{L} = 150 \text{ ab}^{-1} = 1.5 \times 10^8 \text{ pb}^{-1}$, so even a small cross section can produce many events.

Equation (36) is not yet an experimental prediction. It counts events before asking whether the detector can see them or whether they pass the analysis selection. For that reason collider analyses introduce an efficiency factor. In this paper, *signal efficiency* means the fraction of produced signal events that end up contributing to the search histogram after lifetime requirements, detector acceptance, photon reconstruction, angular-resolution requirements, and object-level selections. Schematically,

$$N_{\text{selected}} = \sigma \mathcal{L} \epsilon_{\text{sig}}. \quad (37)$$

The efficiency is not one number fixed by nature. It depends on the ALP mass and coupling. A weakly coupled ALP may live long enough to escape the detector, so it is efficient for the invisible channel and inefficient for the prompt channel. A very light ALP may decay promptly, but its two photons may be so close together that the detector reconstructs them as one merged object. A high-mass ALP may produce well-separated photons, but with a smaller production cross section because the phase space is reduced. The purpose of the analysis is to combine all of these effects consistently.

This distinction also explains why the analytic boundaries in the topology map are not the final exclusion contours. The analytic boundaries answer questions such as “does the ALP decay before the calorimeter?” and “are the two photons separated by more than the angular resolution?” They describe the ideal physics topology. The projected sensitivity contour asks a different question: “would this signal be large enough, after detector reconstruction, to stand out from the Standard Model background?” To answer that, the analytic signal model must be corrected by detector information and compared with simulated background distributions.

The analysis therefore keeps two signal counts separate. The *expected signal yield*, N_{sig} , is the number of ALP signal events predicted for a chosen $(m_a, g_{a\gamma\gamma})$ point after applying the signal probability, efficiency, and detector correction. The *required signal yield*, $N_{\text{sig}}^{\text{req}}$, is the number of signal events that would be needed to exclude the background-only hypothesis at the chosen confidence level. A point is projected to be excluded when

$$N_{\text{sig}}(m_a, g_{a\gamma\gamma}) \geq N_{\text{sig}}^{\text{req}}(m_a). \quad (38)$$

The left-hand side is mostly signal physics. The right-hand side is mostly background statistics. The simulation pipeline exists to compute the pieces of both sides that cannot be obtained reliably from the tree-level analytic formulae alone.

3.2 Signal generation and computational strategy

Signal events are generated using MADGRAPH5_AMC@NLO [13] with a UFO model implementing the ALP effective field theory of Eq. (1). The analysis is deliberately not a pure Monte Carlo scan, and it is also not a purely analytic plot. The reason is that the two approaches answer different questions. The analytic formulae give the ideal production rate, the decay probability, the recoil-photon energy, and the approximate diphoton opening angle. They are excellent for mapping the physics over the full parameter space. For example, at a point $(m_a, g_{a\gamma\gamma})$ one can immediately compute an ideal signal yield of the form

$$N_{\text{ideal}} = \mathcal{L} \sigma(e^+e^- \rightarrow \gamma a) P_{\text{region}}. \quad (39)$$

This already tells us whether the ALP tends to be invisible, prompt, displaced, or merged. It does not, however, tell us everything needed for an experimental sensitivity contour. A detector does not see the analytic final state directly. It sees reconstructed photons with finite angular and energy resolution, finite acceptance, and finite reconstruction efficiency. It may miss photons, merge nearby photons, smear the recoil energy, or move a reconstructed mass peak across a bin boundary. Most importantly, an exclusion contour is not determined only by how many signal events are produced; it is determined by how large that signal would look compared with the Standard Model background in the same observable.

This is why the project uses a hybrid strategy. The analytic calculation gives the signal model over the dense $(m_a, g_{a\gamma\gamma})$ grid. The simulation pipeline is then used only for the parts that cannot be trusted from the analytic formulae alone: the detector-level object reconstruction, the validation of the UFO and lifetime convention, the reconstructed signal observables, and the Standard Model background shapes. The final contour is the place where these two pieces meet: analytic signal production and decay probabilities, corrected by detector-level information, compared against simulated background histograms.

The dense exclusion scan is not produced by running a full detector simulation at every point in the 180×180 mass–coupling grid. Instead, the code uses a two-level strategy:

1. The full $(m_a, g_{a\gamma\gamma})$ grid is evaluated analytically using Eqs. (18)–(34). This gives the production cross section, lifetime, decay-region probability, recoil energy, and opening angle for each point.
2. A finite set of contour-relevant points is passed through the full detector-level chain. These points are used to validate the matrix element, lifetime convention, Delphes output, and analysis observables, and to build branch-aware detector correction factors.

This separation is possible because associated production contains one ALP photon vertex, so at tree level $\sigma(e^+e^- \rightarrow \gamma a) \propto g_{a\gamma\gamma}^2$ exactly. The shape of the production process is therefore controlled mainly by m_a and $\sqrt{s}$, while the coupling rescales the yield and changes the lifetime. This is the main computational shortcut of the project. If the full MadGraph–Pythia–Delphes chain were run for every one of the 32 400 grid points, the campaign would spend most of its time rediscovering quantities that are already known exactly at tree level. The production normalisation changes as $g_{a\gamma\gamma}^2$, the decay length follows from the analytic width, and the resolved/invisible classification follows from comparing ℓ_a and $\Delta\theta_{\gamma\gamma}$ to detector scales. The Monte Carlo is therefore used where it is most valuable: to validate that the UFO really implements the same matrix element, to check that the lifetime handed to Pythia matches the analytic lifetime, to see how the final-state photons appear after detector simulation, and to build correction factors that translate the ideal analytic selection into a reconstructed-object selection.

Concretely, the signal simulation does not define the full contour point by point. Instead, it answers four practical questions. First, does MadGraph give the same production cross section as Eq. (18) for the requested $(m_a, g_{a\gamma\gamma}, \sqrt{s})$? Second, does the width written in the parameter card imply the expected proper lifetime and therefore the expected decay behaviour in Pythia? Third, after Delphes, do prompt-resolved events reconstruct a diphoton invariant mass near m_a , and do invisible events reconstruct a recoil photon near Eq. (11)? Fourth, what fraction of generated events survives the detector-level object selection compared with the ideal analytic selection? The answer to the fourth question is the Delphes correction C_{Delphes} used later in Eq. (52). In this sense, the signal simulation calibrates the analytic contour rather than replacing it.

The grid calculation is explicit. For each logarithmically spaced mass m_a and coupling $g_{a\gamma\gamma}$, the code computes

1. the production cross section from Eq. (18), converted to pb;
2. the recoil-photon energy from Eq. (11);
3. the decay width from Eq. (26);
4. the boosted decay length ℓ_a from Eq. (28);
5. the survival, prompt-decay, and displaced-decay probabilities from Eqs. (30)–(32);
6. the minimum diphoton opening angle from Eq. (33);
7. the detector topology label: invisible, prompt-resolved, displaced, or merged.

This produces the signature map over the full $(m_a, g_{a\gamma\gamma})$ plane. The exclusion contours are then found by solving for the value of $g_{a\gamma\gamma}$ at which the expected signal yield equals the required signal yield at each mass, rather than by simulating every point in the two-dimensional plane.

3.3 Choice of scan ranges

The final FCC-ee projection is evaluated on a logarithmic 180×180 grid,

$$m_a \in [10^{-2}, 80] \text{ GeV}, \quad g_{a\gamma\gamma} \in [10^{-8}, 10^{-1}] \text{ GeV}^{-1}. \quad (40)$$

The lower mass bound, 10 MeV, is chosen because below this value the FCC-ee Z-pole diphoton opening angle $\Delta\theta_{\min} \simeq 4m_a/\sqrt{s}$ is far below the conservative IDEA angular resolution used here. For example, at $m_a = 10 \text{ MeV}$ the opening angle is only 0.025° , compared with $\Delta\theta_{\text{res}} = 1.5^\circ$. Any prompt decay would therefore be a merged-photon object, while weakly coupled ALPs are invisible. Dedicated merged-cluster and very-low-mass beam-dump-style treatments are outside the scope of the current collider projection.

The upper mass bound, 80 GeV, is set by the kinematic and practical reach of associated production at $\sqrt{s} = 91.2$ GeV. The hard kinematic condition is $m_a < \sqrt{s}$, but the recoil photon energy becomes small and the $(1 - m_a^2/s)^3$ phase-space factor suppresses the cross section near the endpoint. Stopping at 80 GeV retains the high-mass collider region where the prompt-resolved channel is meaningful, while avoiding a threshold-dominated tail where a full experimental analysis would need a different low-recoil selection.

The coupling interval in Eq. (40) is deliberately wider than the final visible contours. Values below 10^{-8} GeV^{-1} are far below the 150 ab^{-1} event-yield threshold because $\sigma \propto g_{a\gamma\gamma}^2$. Values up to 10^{-1} GeV^{-1} are included to ensure that the invisible upper lifetime boundary is bracketed for the lightest masses. This upper end should not be interpreted as an equally reliable EFT region for every ALP model. In the UFO parameter-card writer we keep $f_a = 1000 \text{ GeV}$ fixed and vary $K_B + K_W$ to realise the requested physical $g_{a\gamma\gamma}$; very large $g_{a\gamma\gamma}$ therefore corresponds to large dimensionless Wilson coefficients and is used mainly as a numerical bracket for the lifetime transition.

The detector-level chain for one signal point is:

$$\begin{aligned} \text{param_card} &\rightarrow \text{MADGRAPH (LHE)} \rightarrow \text{PYTHIA (HepMC)} \\ &\rightarrow \text{DELPHES (ROOT)} \rightarrow \text{Python validation.} \end{aligned} \tag{41}$$

Each arrow in Eq. (41) has a distinct role. The parameter card sets the physical ALP mass, coupling, and width for one point. MadGraph then computes the tree-level matrix element and writes an LHE event file containing parton-level events and the integrated production weight. Pythia reads the LHE events, adds QED initial- and final-state radiation, decays the ALP according to the supplied width and lifetime, and writes a HepMC event record containing stable final-state particles. Delphes reads the HepMC file and applies a parameterised detector response. Finally, the Python analysis reads the Delphes ROOT file with `uproot/awkward`, reconstructs the analysis observables, and checks that the point passed all validation gates.

The same chain is also used for backgrounds, with one important difference: there is no ALP mass or coupling in the background samples. The background samples are Standard Model processes generated at the same collider energy and passed through the same Pythia and Delphes detector treatment. They provide the expected distribution of ordinary events in the observables where the ALP would be searched for. For the prompt-resolved channel this observable is $M_{\gamma\gamma}$, because the ALP would appear as a narrow peak in a three-photon event. For the invisible channel it is the recoil photon energy, because an invisible ALP would appear as a mono-energetic photon plus missing energy. These background histograms are then used by the limit solver to decide how many signal events would be needed to stand out.

Signal and background are simulated as separate samples rather than as one mixed

dataset. This is standard for a projected search. The signal depends on two continuous parameters, m_a and $g_{a\gamma\gamma}$, so mixing it into the background would require a new full dataset for every possible ALP hypothesis. Instead, the signal shape and efficiency are computed for representative ALP points, and the total signal yield is scaled analytically to any coupling. The background does not depend on m_a or $g_{a\gamma\gamma}$, so each background process only needs to be simulated once at the chosen collider energy and detector setting. The two samples are then combined statistically in the binned limit calculation. This avoids the impossible task of generating a literal FCC-ee dataset with billions of background events for every ALP mass.

The final production inputs are:

- the analytic grid of $180 \times 180 = 32\,400$ mass–coupling points;
- 284 detector-level ALP signal points, each generated with 10 000 events, for a total of 2.84×10^6 generated signal events;
- one $e^+e^- \rightarrow \gamma\gamma\gamma$ background sample with 10 000 generated events;
- one $e^+e^- \rightarrow \gamma\nu\bar{\nu}$ background sample with 10 000 generated events.

The generated event counts are not the luminosity-scaled event counts. They are finite Monte Carlo samples used to estimate shapes and efficiencies. The physical normalisation comes from the MadGraph cross section multiplied by the FCC-ee luminosity.

The imported UFO does not expose $g_{a\gamma\gamma}$ directly. The code writes a point-specific parameter card by mapping the physical coupling to the UFO parameters f_a , K_B , and K_W through

$$g_{a\gamma\gamma} = \frac{\alpha_{\text{em}}(K_B + K_W)}{\sqrt{2}\pi f_a}. \quad (42)$$

The UFO also contains an $a\gamma Z$ vertex. In the imported model this vertex is proportional to

$$C_{\gamma Z} \propto K_W c_w^2 - K_B s_w^2, \quad (43)$$

where s_w and c_w are computed from the same electroweak inputs as the UFO. The default parameter-card writer therefore uses

$$K_B = (K_B + K_W)c_w^2, \quad K_W = (K_B + K_W)s_w^2, \quad (44)$$

which makes Eq. (43) vanish at tree level while keeping the $a\gamma\gamma$ coupling of Eq. (42). This choice is essential for the interpretation of the project. It isolates the one-parameter photophilic benchmark, makes the MadGraph production normalisation comparable to Eq. (18) and to the Belle II closure, and prevents the Z-pole projection from being contaminated by resonant $Z \rightarrow \gamma a$ production. Including a genuine ALP–Z coupling is a

separate physics question with a different operator, different rates, and different signal interpretation.

For lifetime-dependent results, the code writes the project convention

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi} \quad (45)$$

directly into the `DECAY 9999` block of the parameter card. Pythia then decays the ALP to $\gamma\gamma$ with the corresponding proper lifetime. The resulting HepMC event record is passed to DELPHES 3 [15] for fast detector simulation using the IDEA detector card [10].

3.4 Repository implementation and data products

The repository mirrors the physics pipeline. Analytic formulae and validation gates live in the theory layer; MadGraph, Pythia, HepMC, and Delphes wrappers live in the MC layer; background building, signal-efficiency correction, projection solving, and plotting live in the analysis layer. The main machine-readable inputs are the locked JSON files specifying the Belle II closure and FCC-ee Z-pole assumptions. The main checked-in outputs are:

- the analytic theory grid;
- the Belle II closure curve, plot, and JSON summary;
- binned FCC-ee background histograms for the resolved and invisible channels;
- the collected 284-point detector-level ALP signal summary;
- the branch-aware full-analysis efficiency map;
- the FCC-ee projection CSV and signature-classification grid;
- the full, close-up, and combined ALP-photon money plots.

Large raw production files such as LHE, HepMC, ROOT, and full Condor work directories are deliberately not versioned. They are reproducible intermediate products, while the compact CSV, JSON, PNG, and PDF outputs are the report-facing deliverables.

3.5 Detector modelling

The IDEA detector concept [10] is a proposed FCC-ee detector featuring a solenoidal magnetic field, silicon tracking, a drift chamber, and a crystal electromagnetic calorimeter. Delphes is a fast simulation rather than a full GEANT4 detector model. It does not propagate every particle through detailed detector material. Instead, it takes the stable particles in the HepMC event record and applies detector-card modules that approximate the main experimental effects: geometric acceptance, reconstruction efficiency, momentum

and energy smearing, particle-flow object construction, isolation, and missing transverse momentum. For this project the relevant reconstructed objects are photons and missing energy. A photon that falls outside the allowed pseudorapidity range, falls below the energy threshold, or fails the parametrised reconstruction efficiency does not appear as a reconstructed photon in the output ROOT tree. A photon that is reconstructed has its energy and direction smeared according to the detector card. The output is therefore not a private FCC-ee reconstruction, but it is a consistent detector-level approximation that lets the same selection and observable-building logic be applied to signal and background.

This step is conceptually different from the analytic calculation. The analytic calculation knows the truth-level four-vectors and the exact ALP mass used to generate the event. The analysis code is not allowed to use that information when constructing the projected search observable. It uses only reconstructed Delphes objects: the number of photons, their reconstructed energies and directions, and the reconstructed missing-energy object. This distinction is important. The theory can say that a prompt ALP decay produces three photons, but Delphes tells us whether those photons would actually be reconstructed as three photons after detector acceptance and smearing. The theory can say that an invisible ALP gives a recoil photon at a fixed energy, but Delphes tells us how that energy appears in the reconstructed histogram. The final contour uses this reconstructed-object information through the efficiency correction map and through the background histograms.

The key detector parameters used in this analysis are:

- Photon acceptance: $|\eta| < 3.0$, $E_\gamma > 0.5 \text{ GeV}$;
- Photon reconstruction efficiency: 99%;
- Diphoton angular resolution: $\Delta\theta_{\text{res}} = 1.5^\circ$ (conservative calorimeter-cell scale, $\pi/120$);
- Tracker inner radius (L_{min}): 0.02 m;
- Tracker/calorimeter outer radius (L_{max}): 2.5 m.

These values are locked in `analysis/configs/fccee_zpole_inputs.json`. The final deliverable uses the assumptions in Table 1; changing any of these inputs requires regenerating the classification grid, contours, and money plots.

3.6 Validation gates

Before applying the analysis to FCC-ee kinematics, the pipeline is checked by a set of validation gates. These gates are run by the repository validation scripts and by the full signal-production wrapper. They test both the physics formulae and the file handoff between programs.

The Belle II closure is the most important external validation anchor, but it has to be interpreted carefully. Belle II operates at $\sqrt{s} = 10.58 \text{ GeV}$ with 445 pb^{-1} and uses the same process $e^+e^- \rightarrow \gamma a$, $a \rightarrow \gamma\gamma$. However, the public paper does not provide the full binned likelihood, background templates, detector-response maps, or event-level data needed to reproduce the experimental analysis from first principles. The closure test therefore works at the published-contour level. It asks a narrower but still essential question: if we take the public Belle II exclusion boundary as the target, do our cross sections, lifetimes, acceptance cuts, prompt/resolved logic, unit conversions, interpolation, and limit-solving machinery reconstruct that same boundary?

The first step is to load the digitised Belle II contour from AxionLimits [12]. The AxionLimits file stores the excluded region as a polygon in $(m_a, g_{a\gamma\gamma})$, with m_a tabulated in eV. The code converts the masses to GeV, removes the artificial high-coupling cap used to close the filled polygon, and keeps the lower boundary $g_{\text{pub}}(m_a)$ of the published exclusion. This lower boundary is the coupling at which Belle II excludes each ALP mass.

For every point on this boundary, the closure code evaluates our own prompt/resolved event model at the published coupling:

$$N_{\text{pub}}^{\text{model}}(m_a) = \mathcal{L}^{\text{Belle}} \sigma(e^+e^- \rightarrow \gamma a; m_a, g_{\text{pub}}) P_{\text{prompt}}(m_a, g_{\text{pub}}) A_{\gamma}^{\text{Belle}} \Theta_{\text{res}}(m_a) \Theta_E(m_a). \quad (46)$$

Here $\mathcal{L}^{\text{Belle}} = 445 \text{ pb}^{-1}$, $A_{\gamma}^{\text{Belle}}$ is the analytic recoil-photon angular acceptance obtained by integrating the $1 + \cos^2 \theta$ distribution over the Belle II polar-angle range, $P_{\text{prompt}} = 1 - \exp(-L_{\text{min}}/\ell_a)$ is the probability that the ALP decays before the prompt scale, and the two step functions enforce the Belle II photon-energy threshold and the condition that the two photons from $a \rightarrow \gamma\gamma$ are angularly resolved. The photon reconstruction efficiency is set to one in this closure because the unknown Belle II efficiencies, background treatment, and statistical procedure are absorbed into the inferred effective signal threshold. Thus $N_{\text{pub}}^{\text{model}}$ should not be read as Belle II's literal counting threshold; it is the effective number of signal events that our public model would assign to the published boundary.

The closure is then a round-trip test. The code interpolates both $g_{\text{pub}}(m_a)$ and $N_{\text{pub}}^{\text{model}}(m_a)$ in log space. For a dense set of 300 masses it solves

$$N_{\text{prompt}}^{\text{model}}(m_a, g_{\text{closure}}) = N_{\text{pub}}^{\text{model}}(m_a) \quad (47)$$

for g_{closure} using a one-dimensional root finder. If the implementation is internally consistent, the reconstructed $g_{\text{closure}}(m_a)$ should lie on top of the published $g_{\text{pub}}(m_a)$. A mismatch would immediately point to an error in one of the ingredients that enters the event model, such as the GeV/eV conversion, the GeV^{-2} to pb conversion, the factor of 64π in the width, the lifetime-to-decay probability mapping, the angular-resolution cut, or the numerical contour solver.

The closure test yields

$$\max_{m_a} \left| \log_{10} \left(\frac{g_{\text{closure}}}{g_{\text{published}}} \right) \right| = 7.6 \times 10^{-3}, \quad (48)$$

well within the 2% tolerance used as the acceptance criterion. This demonstrates that the public-contour implementation of the analytic model, lifetime convention, and prompt/resolved signal-region logic is internally consistent and agrees with the published Belle II boundary at the level needed for this project. It does not prove that we have reproduced the private Belle II likelihood. Rather, it validates the ingredients that are portable to FCC-ee before we use them in the FCC-ee projection.

3.7 SM background samples

Background events are simulated at $\sqrt{s} = 91.2 \text{ GeV}$ with MADGRAPH5_AMC@NLO + PYTHIA 8 + DELPHES/IDEA. Two channels are generated:

- **Resolved-prompt background:** $e^+e^- \rightarrow \gamma\gamma\gamma$ (SM QED triple-photon production), with $\sigma = 7.306 \text{ pb}$ and $N_{\text{gen}} = 10\,000$ events. After Delphes and object selection, the resolved histogram contains 23 592 diphoton-pair entries. This number can be larger than the number of selected events because one event with several photons contributes several photon pairs.
- **Invisible background:** $e^+e^- \rightarrow \gamma\nu\bar{\nu}$ (radiative neutrino production), with $\sigma = 134.9 \text{ pb}$ and $N_{\text{gen}} = 10\,000$ events. After Delphes and the exactly-one-photon selection, the recoil-energy histogram contains 2 684 entries.

These samples are chosen because they are the irreducible Standard Model versions of the two signal topologies. A prompt-resolved ALP event contains three photons, so ordinary QED three-photon production can fake the same object count and contributes a smooth continuum under the $M_{\gamma\gamma}$ signal peak. An invisible ALP event contains one photon and missing energy, so radiative neutrino production gives the same reconstructed topology without any ALP. Reducible backgrounds, such as misidentified particles or machine-induced activity, are not included in the present projection and are listed as limitations.

The two simulated backgrounds are treated in exactly the same reconstruction language as the signal. For the resolved channel, the Delphes ROOT file is scanned for events with at least three reconstructed photons, and all reconstructed photon pairs in those events are converted into diphoton invariant masses. This produces the background shape in the same variable used to search for the ALP mass peak, $M_{\gamma\gamma}$. For the invisible channel, the ROOT file is scanned for events with exactly one reconstructed photon, and

the photon energy is used as the recoil-energy observable. This is the same variable that maps to the ALP mass through Eq. (11).

The background is not added to the signal event sample event-by-event. Instead, it is added statistically as a binned expectation. For each channel, the background observable values are histogrammed into bins and normalised to the target FCC-ee luminosity,

$$B_i = n_i^{\text{raw}} \frac{\sigma_{\text{bkg}} \mathcal{L}}{N_{\text{gen}}}, \quad (49)$$

where n_i^{raw} is the number of simulated background entries in bin i , σ_{bkg} is the MadGraph cross section for the background process, N_{gen} is the number of generated background events, and $\mathcal{L}$ is expressed in inverse pb. These binned values B_i are what enter the limit calculation. At full luminosity, the expected number of background events across the analysis histograms is 2.58×10^9 (resolved) and 5.43×10^9 (invisible). The resolved background is the dominant QED irreducible process for the three-photon topology; the invisible background is the leading electroweak process with a photon plus missing energy.

The role of the background is therefore very specific. It does not change the ALP production rate, lifetime, or signature classification. Those are signal properties. The background changes the amount of signal required for a point to be excluded. If the background in the relevant mass or recoil-energy bin is large, the signal must be larger to produce the same statistical tension. If the background is small, the analysis approaches the conservative few-event limit. This is why the final contour cannot be obtained by plotting only the analytic signal regions: the analytic regions say what kind of event the ALP would make, while the background histograms say how difficult that event is to distinguish from the Standard Model.

4 Analysis

4.1 Signature classification grid

Each of the $180 \times 180 = 32\,400$ grid points $(m_a, g_{a\gamma\gamma})$ is classified into one of the four signatures using the analytic formulae of Section 2. This classification is not a statistical limit yet; it is a physics map of what the detector would see if an ALP with that mass and coupling were produced. The code first computes the mean lab-frame decay length ℓ_a and the minimum photon separation $\Delta\theta_{\text{min}}$ for the point. It then compares these values to the detector scales L_{min} , L_{max} , and $\Delta\theta_{\text{res}}$. If $\ell_a > L_{\text{max}}$ the ALP is labelled invisible. If it decays before L_{max} but $\Delta\theta_{\text{min}} < \Delta\theta_{\text{res}}$, the photons are labelled merged. If the photons are resolvable and the decay happens before L_{min} , the point is prompt-resolved. If the photons are resolvable but the decay happens between the two length scales, the point is displaced-resolved. This grid is useful because it shows which experimental analysis would

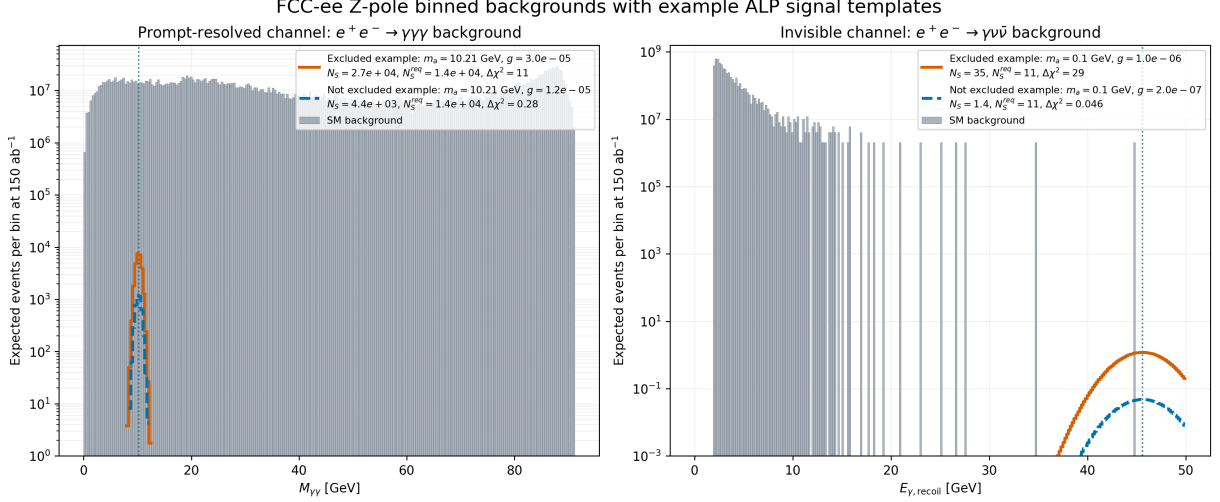


Figure 2: Binned Standard Model background histograms used in the FCC-ee limit calculation, with example ALP signal templates overlaid. Left: the prompt-resolved channel uses the diphoton invariant mass $M_{\gamma\gamma}$ in reconstructed three-photon events. Right: the invisible channel uses the recoil-photon energy. Orange curves show example ALP hypotheses that pass the $\Delta\chi^2 = 2.71$ 90% CL exclusion criterion; blue dashed curves show hypotheses with the same mass but smaller coupling that do not pass the criterion. The signal curves are plotted as expected selected signal events per bin, while the grey histograms show the luminosity-normalised SM background.

be appropriate before any sensitivity calculation is performed. With the IDEA detector assumptions, the grid decomposes as follows (Table 3):

The prompt-resolved region dominates at high $g_{a\gamma\gamma}$ and intermediate m_a , where the ALP decays promptly and the photons are resolvable. The invisible region covers most of the low-coupling parameter space, where the decay length exceeds the detector radius. The prompt-resolved threshold mass is set by $\Delta\theta_{\min} = \Delta\theta_{\text{res}}$:

$$m_a^{\text{thresh}} = \frac{\sqrt{s} \Delta\theta_{\text{res}}}{4} \approx \frac{91.2 \text{ GeV} \times 0.0262}{4} \approx 0.597 \text{ GeV}. \quad (50)$$

Below this mass, the photon pair from ALP decay would be reconstructed as a single cluster, falling into the merged category.

4.2 Signal efficiency and detector corrections

In an ideal analytic calculation, every produced signal event that satisfies the lifetime condition would be counted. A real analysis cannot do that. The event must leave reconstructable detector objects, and those objects must pass the analysis selection. The signal efficiency is the fraction that survives this chain:

$$\epsilon_{\text{sig}} = \frac{\text{number of produced signal events that enter the search histogram}}{\text{number of produced signal events}}. \quad (51)$$

In practice this efficiency is built from two parts. The first part is parametric and simple enough to write down: geometric acceptance, photon efficiency, and the probability that the ALP decays in the desired detector region. The second part is detector-specific and is estimated with Delphes: smearing, object reconstruction, bin migration, and the detailed effect of the analysis selection on reconstructed photons.

The expected number of signal events for a given grid point is:

$$N_{\text{sig}} = \mathcal{L} \cdot \sigma(m_a, g_{a\gamma\gamma}) \cdot P_{\text{region}}(m_a, g_{a\gamma\gamma}) \cdot \epsilon_{\text{param}} \cdot C_{\text{Delphes}}, \quad (52)$$

This equation is the central bridge between theory and the projected contour. The integrated luminosity $\mathcal{L}$ sets the total exposure. The production cross section $\sigma(m_a, g_{a\gamma\gamma})$ determines how many ALPs are produced before any detector or lifetime requirement. The factor P_{region} is the probability that the produced ALP falls into the target signature region:

$$P_{\text{invisible}} = e^{-L_{\text{max}}/\ell_a}, \quad (53)$$

$$P_{\text{prompt}} = 1 - e^{-L_{\text{min}}/\ell_a}. \quad (54)$$

The parametric efficiency ϵ_{param} accounts for the simplest detector effects that can be written down analytically: angular acceptance and photon reconstruction efficiency. The angular acceptance is computed by integrating the production angular distribution $1 + \cos^2 \theta$ over the detector pseudorapidity range $|\eta| < 3.0$. For the invisible channel, the event needs one reconstructed photon, so the efficiency is this angular acceptance times one photon-efficiency factor. For the prompt-resolved channel, the event needs the recoil photon and two ALP-decay photons, so the same acceptance is multiplied by three photon-efficiency factors. The final factor, C_{Delphes} , corrects this idealised parametric estimate using detector-level signal samples.

The reason for this correction is that the analytic boundaries are sharp, while the detector response is not. Analytically, a point is either above or below an angular-resolution threshold, either inside or outside an energy requirement, and either prompt or not prompt according to a mean decay length. In a reconstructed sample, photons are smeared, photons near a threshold can migrate between bins, and finite reconstruction efficiency means that an ideal three-photon event may appear with only two reconstructed photons. Without a detector correction, the contour would be the sensitivity of an ideal detector, not of the IDEA-like detector model used in this project.

The detector correction factor C_{Delphes} is derived from a channel-aware scan of 284 signal points through the full DELPHES simulation. These points are chosen from the contour-relevant branches: the invisible lower branch, the invisible upper branch, and the prompt-resolved branch. For each point, the code reads the Delphes ROOT file and

applies the same object-level logic used in the analysis. In the invisible channel it asks how many generated events become exactly one reconstructed photon in the recoil-energy histogram. In the prompt-resolved channel it asks how many generated events become at least three reconstructed photons with a diphoton mass entering the analysis bins. The measured detector-level selected fraction is then compared to the flat analytic selected fraction $P_{\text{region}}\epsilon_{\text{param}}$. Their ratio defines the correction,

$$C_{\text{Delphes}} = \frac{f_{\text{selected}}^{\text{Delphes}}}{P_{\text{region}}\epsilon_{\text{param}}}. \quad (55)$$

This ratio has a direct interpretation. If $C_{\text{Delphes}} = 1$, the analytic efficiency already predicts the reconstructed selected fraction. If $C_{\text{Delphes}} < 1$, the analytic estimate was too optimistic and the signal yield is reduced. If $C_{\text{Delphes}} > 1$, the analytic estimate was too pessimistic for that branch, usually because the reconstructed selection keeps events in a smoother way than the sharp analytic boundary. The correction is not fitted to the background or to the final contour by hand; it is computed from the signal simulation and then interpolated in mass. The correction is interpolated in $\log m_a$ separately for each contour branch and then inserted back into Eq. (52). In this way the final contour is still fast to compute, but it carries detector-level information from the full simulation campaign. Across the signal scan, the mean detector correction factors are:

- Invisible lower branch: $\langle C \rangle = 0.998$ (range 0.969–1.003),
- Invisible upper branch: $\langle C \rangle = 7.8 \times 10^6$ (extremely large correction from the non-monotonic g dependence of the invisible yield; this branch is numerically fragile),
- Prompt-resolved: $\langle C \rangle = 1.02$ (range 0.900–2.62).

4.3 Limit-setting method

A projected 90% confidence level (CL) exclusion is derived using a binned Asimov $\Delta\chi^2$ test [16]. The logic is the same in both channels. At a fixed ALP mass the signal is expected to appear in a known place: near $M_{\gamma\gamma} = m_a$ for prompt-resolved decays, or near the recoil energy $E_\gamma = (s - m_a^2)/(2\sqrt{s})$ for invisible decays. The background simulation tells us how many Standard Model events are expected in the neighbouring bins of that same observable. The statistical question is then: how many signal events would have to be added to the background-only expectation before the background-only hypothesis would be excluded at one-sided 90% CL?

For a given mass hypothesis, the analysis observable is:

- Resolved channel: the diphoton invariant mass $M_{\gamma\gamma}$, with the signal modelled as a Gaussian centred at m_a with width $\sigma_M = \max(0.05 m_a, 0.05 \text{ GeV})$;

- Invisible channel: the recoil photon energy E_γ , with the signal modelled as a Gaussian centred at $(s - m_a^2)/(2\sqrt{s})$ with width $\sigma_E = \max(0.05 E_\gamma, 0.5 \text{ GeV})$. The histogram uses 264 bins over $[0, 50] \text{ GeV}$, extending beyond the nominal $\sqrt{s}/2 = 45.6 \text{ GeV}$ endpoint to retain detector- and ISR-smeared signal events.

The word “Asimov” means that the expected data are taken to be exactly the background expectation. In other words, the projected limit asks: if FCC-ee observed no excess over the Standard Model expectation, how large a signal would be excluded? For each mass hypothesis the signal shape is converted into a normalised bin fraction f_i , with

$$\sum_i f_i = 1. \quad (56)$$

The expected signal in bin i is then $N_{\text{sig}} f_i$, while the expected background in the same bin is B_i from Eq. (49). In the Gaussian approximation used by the projection solver, the signal template is integrated analytically over the bin edges using the error function. In the Delphes efficiency-map step, the same quantity is also computed directly from the detector-level histogram of reconstructed recoil energies or best diphoton masses. This is how the analysis checks that the Gaussian template and the detector-level signal shapes are mutually consistent.

The intuition is the same as in a simple counting experiment. If the expected background in a bin is B , then ordinary statistical fluctuations are of order $\sqrt{B}$. A signal of one event is impressive if B is nearly zero, but it is irrelevant if B is millions of events. Therefore the required signal is not a fixed number everywhere. It depends on the background level under the signal peak. The binned analysis generalises this idea by adding information from all bins near the signal peak, weighting each bin by its background uncertainty.

With background-only Asimov data, the binned test statistic reduces to

$$\Delta\chi^2(N_{\text{sig}}) = \sum_i \frac{(N_{\text{sig}} f_i)^2}{\max(B_i, 1)}. \quad (57)$$

The $\max(B_i, 1)$ denominator is a practical floor used to avoid declaring infinite sensitivity in a bin where the finite Monte Carlo sample contains no background entries. It also makes the projection conservative in sparsely populated bins. A point is excluded at one-sided 90% CL when $\Delta\chi^2 = 2.71$. Solving Eq. (57) for N_{sig} gives the required signal count in Eq. (58).

This is the criterion for saying that the signal is “different enough” from the background in the projection. We do not require the signal to be larger than the total background in the whole dataset. That would be far too strict, because the ALP signal is localised in a specific mass or recoil-energy region. Instead, we ask whether adding the

716 signal template would change the binned distribution enough that the background-only
 717 prediction would be disfavoured at 90% CL.

718 For each mass point, the minimum signal yield required for exclusion is:

$$N_{\text{sig}}^{\text{req}} = \max \left(3, \sqrt{\frac{2.71}{\sum_i f_i^2 / \max(B_i, 1)}} \right), \quad (58)$$

719 where f_i is the normalised signal fraction in bin i , B_i is the expected SM background in
 720 that bin, and 3 is a conservative event-count floor. The threshold $\Delta\chi^2 = 2.71$ corresponds
 721 to a one-sided 90% CL. The contour in the $(m_a, g_{a\gamma\gamma})$ plane is then the locus of points
 722 where $N_{\text{sig}}(m_a, g_{a\gamma\gamma}) = N_{\text{sig}}^{\text{req}}(m_a)$. This equation is the final point where all parts of
 723 the project meet. The left side, N_{sig} , comes from the analytic ALP production rate,
 724 analytic decay probability, simple detector acceptance, and Delphes-derived efficiency
 725 correction. The right side, $N_{\text{sig}}^{\text{req}}$, comes from the simulated and luminosity-normalised
 726 Standard Model background in the relevant analysis observable. If N_{sig} is larger than
 727 $N_{\text{sig}}^{\text{req}}$, that point would be excluded in the projected background-only FCC-ee dataset. If
 728 it is smaller, the point would not be excluded by the channels included in this analysis.
 729 Operationally, the code loops over the 180 mass values. For each mass, it first computes
 730 the required signal count from the corresponding background histogram. It then scans
 731 $g_{a\gamma\gamma}$ logarithmically to find sign changes in

$$F(g_{a\gamma\gamma}) = N_{\text{sig}}(m_a, g_{a\gamma\gamma}) - N_{\text{sig}}^{\text{req}}(m_a), \quad (59)$$

732 and refines each sign change with a Brent root finder. The resulting root is the coupling
 733 value plotted on the contour. This root-finding method is used instead of simply drawing
 734 the nearest grid cell, so the contour is not limited to the discrete 180 coupling values used
 735 for the topology map.

736 Because the invisible signal yield is non-monotonic in $g_{a\gamma\gamma}$ —it peaks at intermediate
 737 coupling and falls off both towards small coupling (fewer events produced) and large
 738 coupling (ALP decays before reaching L_{max})—the invisible contour has two branches: the
 739 *lower branch* (sensitivity floor from statistics) and the *upper branch* (sensitivity ceiling
 740 from lifetime).

5 Results

5.1 Signature map

Figure 3 shows the classification of the $(m_a, g_{a\gamma\gamma})$ plane under the IDEA detector assumptions. The four signatures occupy well-separated regions. The invisible region (blue) dominates at low coupling over a wide mass range; the prompt-resolved region (red) covers the high-coupling, intermediate-mass space above the resolved threshold; the merged region (orange) occupies the low-mass, high-coupling corner where photon pairs are not angularly resolved; and the displaced-resolved band (green) forms a narrow transition strip. The physics boundaries between regions—the $\ell_a = L_{\min}$ and $\ell_a = L_{\max}$ curves computed from Eqs. (26) and (28)—are overlaid as dashed lines and agree with the classification boundaries to machine precision.

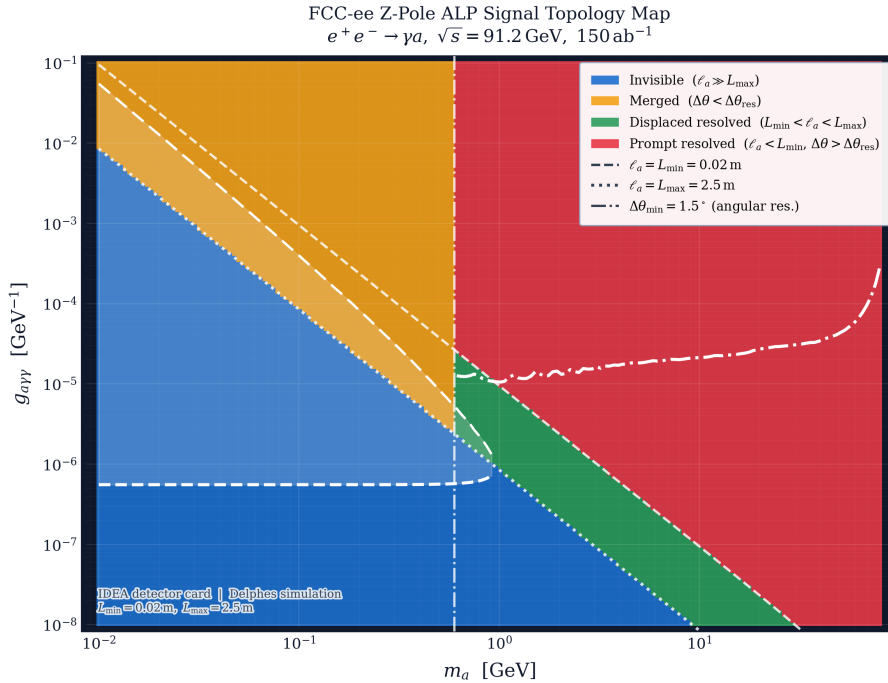


Figure 3: ALP signature classification in the $(m_a, g_{a\gamma\gamma})$ plane for the FCC-ee Z pole ($\sqrt{s} = 91.2 \text{ GeV}$) under IDEA detector assumptions. Dashed lines show the analytic $\ell_a = L_{\min}$ (prompt/displaced boundary) and $\ell_a = L_{\max}$ (displaced/invisible boundary) curves. The thin dash-dot curve shows the angular-resolution boundary $\Delta\theta = \Delta\theta_{\text{res}} = 1.5^\circ$. Only the invisible and prompt-resolved signatures are used in the projected exclusion contours.

5.2 Projected exclusion contours

Figure 4 shows the main result: the projected FCC-ee Z-pole exclusion contours overlaid on the current ALP-photon constraint landscape from AxionLimits [12].

The external regions in Fig. 4 are used as a landscape, not as a single statistically homogeneous exclusion. AxionLimits combines direct laboratory searches, stellar-energy-

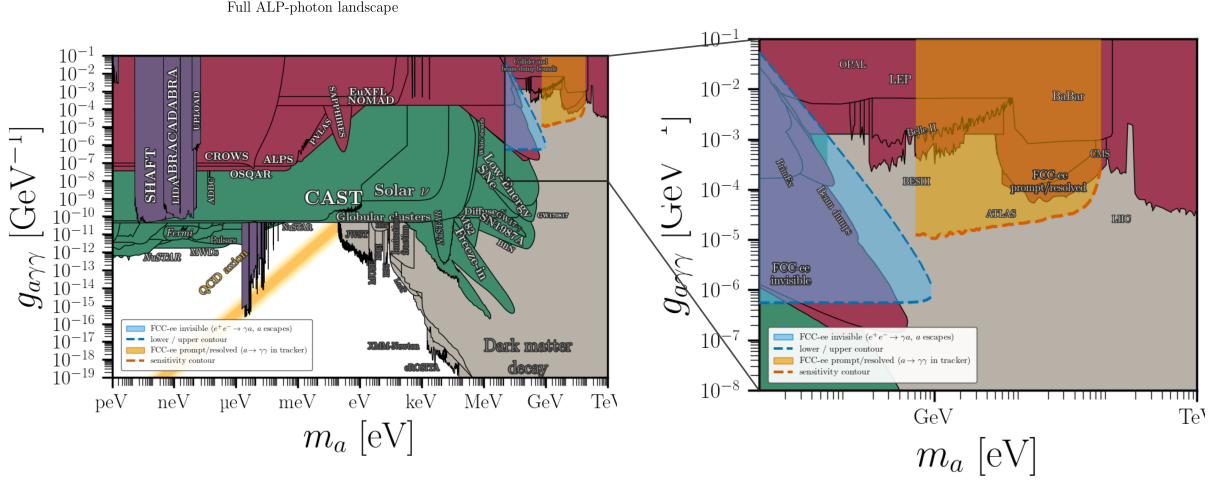


Figure 4: ALP-photon parameter space with FCC-ee Z-pole projected sensitivity. *Left*: full ALP-photon landscape from meV to TeV ALP mass. Individual existing constraints are shown as labelled coloured regions (stellar/astrophysical in green, helioscope CAST in pink, colliders and beam experiments in red/blue tones). *Right*: FCC-ee Z-pole zoom region (10 MeV–80 GeV). Shaded blue band: projected invisible exclusion region, bounded below (solid dashed line, sensitivity floor) and above (lighter dash, lifetime ceiling). Shaded green region: projected prompt-resolved exclusion, bounded by the sensitivity contour (green dashed). All constraints shown are existing published bounds. Background constraints from AxionLimits [12].

loss bounds, cosmological arguments, dark matter-motivated projections, and QCD-axion reference bands. These inputs do not all share the same assumptions about ALP abundance, branching fractions, early-universe history, or statistical treatment. The FCC-ee curves overlaid here are different in character: they are direct collider projections for the photophilic benchmark of Eqs. (1) and (44), with no assumption that the ALP is dark matter or that it solves the strong CP problem. Regions labelled by cosmology, stellar physics, and QCD-axion model lines are therefore important context for the global ALP landscape, but they do not replace the controlled collider question answered by this repository.

Invisible channel. The invisible sensitivity spans $m_a \in [0.01, 0.92]$ GeV. The lower contour (sensitivity floor) sits at:

$$g_{a\gamma\gamma}^{\text{inv}} \approx 5.5 \times 10^{-7} \text{ GeV}^{-1} \quad (\text{nearly flat across the mass range}), \quad (60)$$

representing a factor ~ 10 –100 improvement over the current best collider bounds in this mass range. The near-flatness of the lower contour reflects the dominance of the signal statistics floor (3 events) over the background shape at high luminosity. The upper contour rises from $g_{a\gamma\gamma} \approx 1.3 \times 10^{-6} \text{ GeV}^{-1}$ at $m_a = 0.01 \text{ GeV}$ to $g_{a\gamma\gamma} \approx 0.055 \text{ GeV}^{-1}$ at

$m_a = 0.92 \text{ GeV}$, marking the transition to the short-lifetime (prompt) regime.

Prompt-resolved channel. The prompt-resolved sensitivity spans $m_a \in [0.61, 80] \text{ GeV}$.

The lower mass threshold is set by Eq. (50). The exclusion contour runs from:

$$g_{a\gamma\gamma}^{\text{res}} \in [1.1 \times 10^{-5}, 2.9 \times 10^{-4}] \text{ GeV}^{-1} \quad (\text{at } m_a = 0.61\text{--}80 \text{ GeV}). \quad (61)$$

In the mass range 1–10 GeV, the prompt-resolved sensitivity at FCC-ee extends below the current Belle II and BaBar limits by roughly one to two orders of magnitude.

Comparison to existing limits. Table 4 summarises the FCC-ee reach at selected masses compared to the best current bounds.

6 Discussion

6.1 Physics reach

The FCC-ee Z-pole presents a qualitatively different sensitivity landscape from any existing experiment. At low ALP masses ($m_a \lesssim 0.1 \text{ GeV}$), the invisible channel exploits the huge luminosity to push the sensitivity floor below $g_{a\gamma\gamma} \sim 10^{-6} \text{ GeV}^{-1}$, a regime otherwise accessible only to stellar observations (HB stars, SN 1987A). At intermediate masses (1–10 GeV), the prompt-resolved channel cleanly covers the region between current B -factory limits and LEP. At high masses (10–80 GeV), the prompt-resolved channel overlaps with and significantly improves on LEP and LHC constraints, benefiting from the clean e^+e^- environment and the mono-energetic photon tag.

The invisible channel’s sensitivity has a fundamental lower limit set by the 3-event floor in Eq. (58): at exactly 3 required events and $\mathcal{L} = 150 \text{ ab}^{-1}$, the cross section threshold corresponds to $\sigma_{\text{min}} = 3/(1.5 \times 10^8 \text{ pb}^{-1}) = 2 \times 10^{-8} \text{ pb}$. From Eq. (18), this maps to $g_{a\gamma\gamma} \approx 5 \times 10^{-7} \text{ GeV}^{-1}$, consistent with Eq. (60). A nearly background-free invisible analysis at this luminosity would be statistics-limited. In the implemented binned-background analysis, the background shape raises the required signal yield above three events for some masses (for example to about 10.7 events at the lightest masses), but the result remains driven mainly by the enormous FCC-ee luminosity and the $\sigma \propto g_{a\gamma\gamma}^2$ scaling.

6.2 Confidence in the projection

The results should be interpreted as a detector-corrected phenomenological projection, not as an experimental limit. Within that scope, the confidence is highest where the analysis is controlled by validated analytic formulae and moderate detector corrections, and lower where small lifetime probabilities are amplified by large correction factors.

The most robust headline statements are therefore the existence of strong FCC-ee reach in the invisible lower branch and in the prompt-resolved branch, the location of the resolved threshold near 0.6 GeV, and the qualitative separation of invisible, prompt/resolved, merged, and displaced regions. The least robust feature is the precise position of the invisible upper branch, which should be described as the expected lifetime boundary rather than as a precision contour.

6.3 Limitations and caveats

Several important caveats apply to the current projections.

Invisible upper branch. The invisible upper contour (the lifetime ceiling) carries very large detector correction factors ($C_{\text{Delphes}} \sim 10^6\text{--}10^8$ in parts of the low-mass tail). These arise because the analytic invisible probability $P_{\text{inv}} = e^{-L_{\text{max}}/\ell_a}$ plunges steeply as $g_{a\gamma\gamma}$ increases, while the Delphes simulation shows a slightly smoother falloff due to ISR and finite detector response. The upper branch should be treated as directional rather than precise.

Detector systematics. The current analysis includes only the dominant SM backgrounds for each channel and does not account for: systematic uncertainties on photon reconstruction efficiency and energy scale; machine-induced backgrounds from beamstrahlung and synchrotron radiation (relevant at FCC-ee); pileup-like machine backgrounds; and higher-order QED and electroweak corrections to the SM background rates.

Merged and displaced signatures. The merged and displaced-resolved signatures together cover 24% of the $(m_a, g_{a\gamma\gamma})$ grid. Merged photons require dedicated cluster-shape reconstruction, and displaced diphotons require dedicated vertex-level analysis. Including these channels would extend the claimed reach to couplings where the ALP is not fully prompt, potentially filling in the gap between the invisible and prompt-resolved contours.

Single-iteration Delphes correction. The branch-aware detector correction factor is derived from signal points on the first-iteration projected contour, not at the corrected contour positions. A second Delphes campaign at the corrected contour points would be needed for a fully self-consistent result.

Belle II closure scope. The closure test validates the analysis model at published-contour level. It does not access Belle II’s private likelihood, background spectra, or efficiency maps, and should not be read as a full reimplementation of the Belle II result.

6.4 Outlook

Several extensions would strengthen and broaden the analysis.

Displaced and merged channels. Implementing dedicated reconstruction for displaced and merged topologies would extend the FCC-ee ALP coverage to a nearly complete $(m_a, g_{a\gamma\gamma})$ map from the resolved threshold ($m_a \approx 0.6$ GeV) all the way to the kinematic endpoint ($m_a \lesssim \sqrt{s}/2 = 45.6$ GeV), including the intermediate-lifetime region currently left blank.

Higher-order corrections. ISR corrections and NLO QCD/EW effects on the background processes could shift the background normalisation by $\mathcal{O}(10\%)$, which at this level of signal sensitivity may become relevant.

ALP- Z coupling. The present analysis considers only the ALP-photon coupling $g_{a\gamma\gamma}$. ALPs may also couple to the Z boson, modifying both production and decay at FCC-ee energies [5, 6]. In this repository that coupling is cancelled at tree level by Eq. (44). This is why the projection does not use the potentially resonant $Z \rightarrow \gamma a$ enhancement from the 5×10^{12} Z bosons expected at the FCC-ee Z pole. Including a nonzero $C_{\gamma Z}$ would enrich the phenomenology, but it would be a different two-coupling analysis with additional processes such as $Z \rightarrow \gamma a$ and possibly $e^+e^- \rightarrow Za$.

Other FCC-ee run stages. FCC-ee will also operate at the WW threshold, HZ threshold, and $t\bar{t}$ threshold. The WW and HZ stages provide different kinematics and potentially larger cross sections for ALPs coupling to W bosons or via Higgs loops, while the higher $\sqrt{s}$ extends the prompt-resolved reach to higher m_a .

7 Conclusion

We have computed projected 90% CL exclusion contours for photophilic ALPs at the FCC-ee Z -pole ($\sqrt{s} = 91.2$ GeV, $\mathcal{L} = 150$ ab $^{-1}$) using a complete simulation chain from matrix-element generation to fast detector response. For the invisible channel, the analysis achieves sensitivity to $g_{a\gamma\gamma} \approx 5.5 \times 10^{-7}$ GeV $^{-1}$ across the mass range $m_a \in [0.01, 0.92]$ GeV, surpassing current collider bounds by one to two orders of magnitude. For the prompt-resolved channel, the contour covers $m_a \in [0.61, 80]$ GeV at $g_{a\gamma\gamma} \sim 10^{-5}$ – 3×10^{-4} GeV $^{-1}$, significantly exceeding the reach of existing e^+e^- and LHC experiments.

The analysis is validated against the published Belle II exclusion boundary to 0.8%, confirming the consistency of the analytic production model, lifetime convention, and signal-region logic. The simulation chain uses MadGraph5, Pythia 8, and a Delphes IDEA card; the validation suite also checks the MadGraph cross section, the ALP width and Pythia lifetime, the Delphes output, and the invariant-mass or recoil-energy observable for every detector-level signal point. The most robust parts of the result are the invisible

lower branch and the prompt-resolved contour. The invisible upper branch should be interpreted more qualitatively because it is a rapidly varying lifetime boundary with large detector-correction factors.

The results establish FCC-ee as an extraordinarily powerful probe of photophilic ALPs in the 10 MeV–80 GeV mass range, potentially surpassing the world-leading sensitivity of stellar and astrophysical bounds in a controlled collider environment. Future work should extend the analysis to displaced and merged signatures, include machine-induced backgrounds and systematic uncertainties, and incorporate ALP couplings beyond the photon vertex.

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Appendix: Contribution Statement

Iñaki Pardo led the physics study, repository organization, simulation campaigns, analysis development, validation, plotting, and writing. The project was developed in collaboration with project partners and with Codex-assisted implementation and documentation work. External software, models, detector cards, constraint data, and papers are credited in the repository provenance report and bibliography.

Table 1: Locked assumptions used for the final FCC-ee Z-pole deliverable.

Ingredient	Value	Role in the analysis
Collider stage	$\sqrt{s} = 91.2 \text{ GeV}$, $\mathcal{L} = 150 \text{ ab}^{-1}$	Baseline FCC-ee Z-pole scenario; sets recoil energy, production phase space, and event normalisation.
Physics benchmark	Photophilic ALP, $C_{\gamma Z} = 0$ at tree level	Keeps the result a one-coupling $g_{a\gamma\gamma}$ projection rather than a mixed photon-Z benchmark.
Scan domain	$m_a = 0.01\text{--}80 \text{ GeV}$, $g_{a\gamma\gamma} = 10^{-8}\text{--}10^{-1} \text{ GeV}^{-1}$	Covers the FCC-ee mass reach and brackets both the lower sensitivity floor and upper invisible lifetime boundary.
Signal regions	Invisible and prompt-resolved only	These are the two channels with simple robust observables: recoil-photon energy and $M_{\gamma\gamma}$.
Detector lengths	$L_{\min} = 0.02 \text{ m}$, $L_{\max} = 2.5 \text{ m}$	Define prompt, displaced, and invisible decay probabilities.
Photon reconstruction	$ \eta < 3.0$, $E_\gamma > 0.5 \text{ GeV}$, $\epsilon_\gamma = 0.99$	Implements the IDEA-like photon acceptance and reconstruction efficiency.
Angular resolution	$\Delta\theta_{\text{res}} = 1.5^\circ$	Defines whether $a \rightarrow \gamma\gamma$ is resolved or merged.
Analysis resolutions	$\sigma_M = \max(0.05m_a, 0.05 \text{ GeV})$, $\sigma_E = \max(0.05E_\gamma, 0.5 \text{ GeV})$	Widths of the Gaussian signal templates in the resolved and invisible binned likelihoods.
SM backgrounds	$e^+e^- \rightarrow \gamma\gamma\gamma$ and $e^+e^- \rightarrow \gamma\nu\bar{\nu}$	Irreducible backgrounds for the prompt-resolved and invisible channels, respectively.
Statistical criterion	One-sided 90% CL, $\Delta\chi^2 = 2.71$, with a 3-event floor	Converts binned signal-plus-background expectations into exclusion contours.
External constraints	AxionLimits commit 7d375f4879b32406 a239fe48d2615a4b fd9bc0bb	Provides the published ALP, QCD-axion, astrophysical, cosmological, and laboratory landscape used only as context for the money plots.

Table 2: Validation gates implemented in the analysis pipeline.

Gate	What is tested	Current status
Software smoke test	A generic $e^+e^- \rightarrow \mu^+\mu^-$ chain proves that MadGraph, Pythia, HepMC, Delphes, ROOT, and the Python reader work together.	Integrated as the cluster setup test.
Gate 1: cross section	The MadGraph integrated weight for $e^+e^- \rightarrow \gamma a$ is compared to Eq. (18) at the generated $(m_a, g_{a\gamma\gamma}, \sqrt{s})$ point.	Passed for 284/284 detector-level ALP scan points.
Gate 2: width and lifetime	The UFO width convention is checked, the project 64π width is written into the parameter card, and the Pythia input lifetime is checked against Eq. (28).	Convention locked to Eq. (26); Pythia summaries are checked per point.
Gate 3: Belle II closure	The Belle II public contour is reconstructed using the same production, lifetime, and prompt/resolved logic as the FCC-ee projection.	Passed with maximum log residual 7.6×10^{-3} .
Detector-observable gate	Resolved points must reconstruct a diphoton mass near m_a ; invisible points must reconstruct the recoil photon near Eq. (11).	Passed for 284/284 detector-level ALP scan points.
Background gate	SM background ROOT outputs are converted to binned luminosity-normalised histograms used directly by the limit solver.	Two full-stat 10 000-event background channels included.

 Table 3: Signature classification of the $(m_a, g_{a\gamma\gamma})$ grid.

Signature	Grid points	Fraction
Prompt-resolved	14 171	43.7%
Invisible	10 452	32.3%
Merged	5 989	18.5%
Displaced-resolved	1 788	5.5%
Total	32 400	100%

Table 4: FCC-ee projected sensitivity versus current best limits at representative masses. Current limits from AxionLimits [12].

m_a	Channel	Current best $g_{a\gamma\gamma}$ [GeV $^{-1}$]	FCC-ee projected $g_{a\gamma\gamma}$ [GeV $^{-1}$]
0.1 GeV	Invisible	$\sim 10^{-4}$ (beam dumps)	5.5×10^{-7}
1 GeV	Prompt-resolved	$\sim 10^{-3}$ (Belle II)	$\sim 3 \times 10^{-5}$
10 GeV	Prompt-resolved	$\sim 10^{-4}$ (BaBar)	$\sim 2 \times 10^{-5}$
50 GeV	Prompt-resolved	$\sim 10^{-3}$ (LEP)	$\sim 2 \times 10^{-4}$

Table 5: Reliability assessment of the main ingredients in the projection.

Ingredient	Confidence	Reason
Production normalisation	High	The associated-production cross section is analytic and Gate 1 agrees with MadGraph for all 284 detector-level signal points.
Width and lifetime convention	High	The 64π width convention is locked in the parameter-card writer, Pythia receives the same lifetime, and Gate 2 checks the convention.
Belle II closure logic	High at public-contour level	Gate 3 reproduces the digitised Belle II boundary to a maximum residual of 7.6×10^{-3} in $\log_{10} g_{a\gamma\gamma}$.
Prompt-resolved FCC-ee contour	Moderate to high	The branch has reasonable Delphes correction factors (median close to one, range 0.90–2.62) and a direct invariant-mass validation at detector level.
Invisible lower contour	Moderate to high	The lower branch is controlled by production rate, survival probability near unity, recoil-energy validation, and correction factors close to one.
Invisible upper contour	Low to moderate	This is a lifetime ceiling where the invisible probability changes extremely rapidly with $g_{a\gamma\gamma}$; the detector-correction factors become very large in the low-mass tail.
Absolute experimental reach	Moderate	The study includes leading SM backgrounds and Delphes response, but not full detector systematics, machine backgrounds, or a real FCC-ee experimental likelihood.